



Co-occurrence of multidrug resistance, β -lactamase and plasmid mediated AmpC genes in bacteria isolated from river Ganga, northern India[☆]

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ABSTRACT

Wastewater effluents released in surface water provides suitable nutrient rich environment for the growth and proliferation of antibiotic resistant bacteria (ARB) and genes (ARG). Consequently, bacterial resistance has highly evolved over the recent years and diversified that each antibiotic class is inhibited by a distinct mechanism. In the present study, the prevalence of Multidrug resistant (MDR), extended spectrum β -lactamases (ESBL) and plasmid mediated Amp-C producing strains was analyzed in 28 surface water samples collected near domestic effluent discharge sites in river Ganga located across 11 different geographical indices of Uttar Pradesh, India. A total of 243 bacterial strains with different phenotypes were isolated. Among 243 isolates, 206 (84.77%) exhibited MDR trait displaying maximum resistance towards β -lactams (P = 78.19%; AMX = 72.84%), glycopeptides (VAN = 32.92%; TEI = 79.42%), cephalosporins (CF = 67.90%; CFX = 38.27%), and lincosamides (CD = 78.18%) followed by sulfonamide, macrolide and tetracycline. ESBL production was confirmed in 126 (51.85%) isolates that harbored the genes: *bla*TEM (95.24%), *bla*SHV (22.22%), *bla*OXA (11.90%) and *bla*CTX-M group (14.28%). The presence of plasmid mediated AmpC was detected only in 6.17% of isolates. The existence of such pathogenic strains in the open environment generates an urgent need for incorporating stringent measures to reduce the antibiotic consumption and hence its release.

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1. Introduction

The rapid advancement in development (urbanization and industrialization) has not only affected the environment but has also threatened the living beings. The discharges of effluents in aquatic environment/wastewater are responsible for development

of antibiotics resistance in bacteria and other microbes through selection pressure as well as uptakes of antibiotics by aquatic organisms and crops irrigated with polluted water, thus entering food chain and human gut. Antibiotic resistance is an outcome of such unregulated activities and the phenomenon has diversified into various types and sub-types that are predominant throughout the world. Waste water from industries, municipal and hospitals are loaded with diverse range of antibiotics (amoxicillin, doxycycline, cephalixin, ciprofloxacin, etc.), antibiotic resistant bacteria (Kumar and Pal, 2018) as well as many chemical and organic pollutants that are disposed of without any effective treatment into the rivers (Guo et al., 2017; Wellington et al., 2013) thereby generating a co-selection pressure for development of resistance (Karkman et al., 2018). Such effluents serve as nutrient-rich reservoirs (hotspots) for the easy mushrooming (vertical and horizontal gene transfer) of the resistant bacteria in air, water and soil domains of environment (Wellcome Trust, 2018; Ferro et al., 2017; Hultman et al., 2018; Sabri

Abbreviations: MDR, Multidrug Resistant; ESBL, Extended Spectrum Beta Lactamase; ARB, Antibiotic Resistant Bacteria; ARG, Antibiotic Resistant Genes; BOD, Biochemical Oxygen Demand; COD, Chemical Oxygen Demand; TSS, Total Suspended Solid; HGT, Horizontal Gene Transfer; WWTPs, Wastewater Treatment Plants.

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et al., 2018). Due to the inadequate treatment the resistant strains persists in these effluents and are released as such into the rivers or surface waters (Yu et al., 2018; Hembach et al., 2017). These anthropogenic activities have further accelerated the emergence, dissemination and proliferation of the antibiotic resistant bacteria along with the genes in the water ecosystems (Pruden, 2014; Rodríguez-Mozaz et al., 2014; Karkman et al., 2018; Ortega-Paredes et al., 2019). Furthermore, lack of stringent policies on antibiotics usage promotes the overuse and misuse of the same in agricultural and veterinary applications that has accelerated the resistance threat (Berendonk et al., 2015; Sharma et al., 2016). International organizations such as World Health Organization (WHO) and Center for Disease Control and Prevention (CDC) have individually reported the catastrophic effects and status of this outbreak. As per the reports, more than 2.8 million antibiotic-resistant infections affect the lives in United States annually and >35,000 people die as a consequence (CDC, 2019). According to WHO, presently it has affected nearly 7,00,000 people per annum and by 2050 this pandemic could lead to 10 million deaths each year (WHO, 2019). With the onset of this threat many conventional infections related to respiratory, urinary tract, etc are becoming difficult to treat (WHO, 2019; Ram and Kumar, 2020; Kumar et al., 2019). The long-term presence of ARB and ARGs will further be responsible for the relocation, transformation and dispersal of the trait to bacterial species of environmental origin (Gao et al., 2017).

Aquatic systems such as rivers, lakes often receives wastewater from the treatment plants and sometimes raw wastewater as a result of sewer overflow during heavy rainfall (Liu et al., 2020) and thus, provide a suitable platform for the proliferation and dissemination of such pathogenic bacteria. River Ganga holds a religious status and has socio-economic importance in India. Being a perennial river, Ganga and its tributaries are the main source of water for domestic as well as irrigation purposes for the North Indian states. Due to its cultural, spiritual and mythological significance, the river is worshipped as a living goddess in India. The

world's largest religious and cultural human congregation, 'Kumbh' is also organized where three rivers Ganga, Yamuna and Saraswati meet in Prayagraj, India and people from across the globe take holy dip in its flowing water (Kumar, 2017). The river confers various applications in different sectors but due to large-scale anthropogenic activities such as discharge of untreated and unmonitored domestic and industrial effluents; the physico-chemical and bacteriological quality of the river water has been drastically affected in the recent years. The bacterial species thriving in such contaminated environments has adapted itself to the repeated concentrations of the drug by generating distinct mechanisms to withstand its bactericidal effects. The residing microbial population in such areas may develop resistance to antibiotics belonging to different classes and/or may evolve mechanisms that include production of specific enzymes that inactivates the antibiotics. Beta-lactamases are one such class of enzymes that play a significant role in antibiotic resistance. In India, higher resistance against β -lactam antibiotics may be attributed to higher consumption of β -lactam antibiotics (Klein et al., 2018). As the β -lactam antibiotics are broad spectrum and hence widely used in hospitals to treat the bacterial infections. The discharged effluents without improper treatment contains rich concentration of the drug, which usually end up into the rivers through WWTPs (Ye et al., 2017; Rodríguez-Mozaz et al., 2014). The selection pressure to survive works upon thereby creating these reservoirs or aquifers as hot spot of acquired (plasmidic AmpC) or inherent resistance.

These hydrolytic enzymes function by cleaving the β -lactam ring of the antibiotics belonging to penicillins, cephalosporins, carbapenems and monobactams (Bush and Jacoby, 2010; Nnadozie and Odume, 2019). The ESBL/AmpC genes may either be chromosomal or plasmid-mediated. The most prevailing ESBL genotypes that are mainly detected in such environments include *bla*TEM, *bla*SHV, *bla*OXA and *bla*CTX-M (Dallenne et al., 2010) whereas among the AmpC types, the most common plasmid-mediated genes are MOX, FOX, CMY, EBC, DHA and ACC (Munita and Arias, 2016; Ben Said

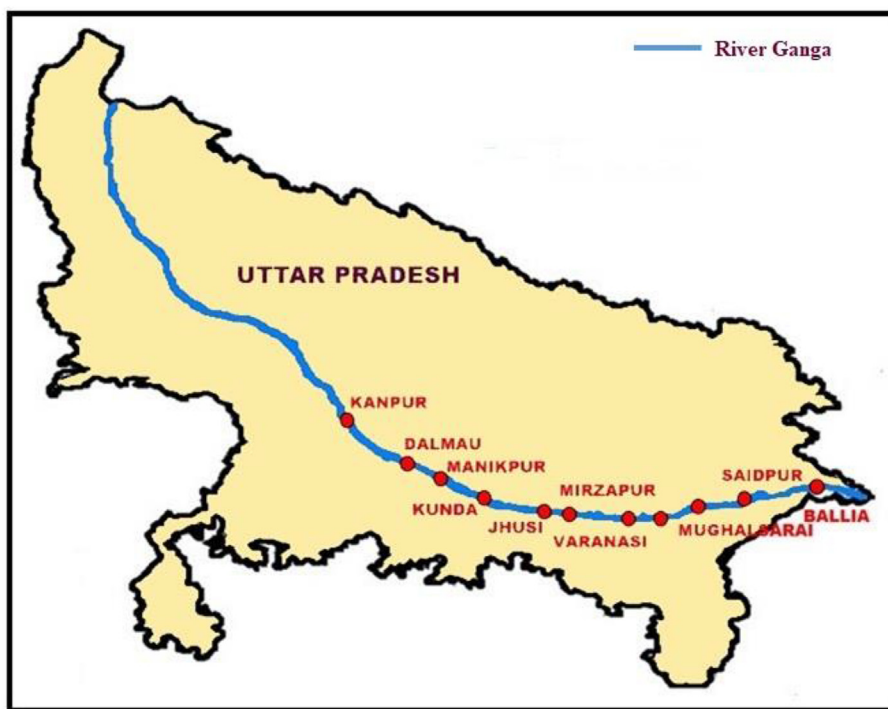


Fig. 1. Sampling stations across river Ganga in Uttar Pradesh, India (Map not to scale).

et al., 2016). The study for the detection of resistant genes is majorly limited within members belonging to Enterobacteriaceae family (*E.coli*, *Klebsiella*, *Shigella*, *Salmonella* etc.) but at present these genes have now also been detected in members of non-clinical environmental species. The resistant bacteria pose major risk to the living population as the treatment options available for such infections are limited (Canton et al., 2012). Besides, natural environments like aquifers, rivers, and water reservoirs etc. acts as a hub for the dissemination of ARG thus aiding in mounting strategy for tracking resistance threat (Makowska et al., 2020). The novelty this study confers: (i) co-existence of ESBL, AmpC and antibiotic resistance in riverine system; (ii) a comparative analysis for variation in multidrug resistance rate in temperate geographical place like India; (iii) multidrug resistance in environmental isolates.

Thus, keeping in view the cultural and socio-economic significance of river Ganga and the increasing incidence of antibiotic resistance in rivers, the present study was conducted with the following objectives: i) to investigate the abundance of MDR, ESBL and/or AmpC harboring bacteria in water samples collected from river Ganga, ii) to detect the co-occurrence of beta-lactamase production and multidrug resistance, iii) molecular detection of the beta-lactamase genes, and iii) characterization and identification of the isolates that possessed the three traits.

2. Material and methods

2.1. Sample collection and isolation of bacterial colonies

The surface water samples were collected near sites based on human intervention and discharge of industrial, hospital and municipal wastes in river Ganga located across 11 different geographical indices of Uttar Pradesh, India (as detailed in Fig. 1, Table 1). Sampling was done during December 2018–June 2019. Grab samples were collected from each sites in sterile wide-mouth glass and plastic bottles. The samples were stored at 4 °C until

processed.

Microbiological analysis was carried out within the first 24 h after the arrival of samples in the laboratory. The heterotrophic bacterial colonies were isolated from the water samples by culture dependent method (Method no. 9215, APHA, 2012). Serial dilutions up to 10^{-6} was made using sterile n-saline (0.9% NaCl) and aliquots of 100 µL from each dilution were spread on Plate Count agar (HiMedia Laboratories, Pvt. Ltd. Mumbai, India) followed by incubation at 37 °C (± 2) for 18–24 h. The isolates were further characterized on the basis of their cell membrane features (Gram staining).

2.2. Analysis of water quality parameters

The collected samples were analyzed based on various water quality parameters using validated and approved methods by APHA, 2012 and Bureau of Indian Standards BIS, 2012. The tests for physico-chemical parameters: pH, color, Biochemical Oxygen Demand (BOD), COD (Chemical Oxygen Demand), and Total Suspended Solid (TSS) were conducted in the laboratory whereas Dissolved Oxygen (DO) was measured on site using a calibrated multi-parameter meter with probes. Simultaneously, bacteriological analysis such as total and fecal coliform was performed using multiple tube fermentation method (Method no. 9221, APHA, 2012). The enumeration of total coliform along with differential count of *E. coli* was also done using HiTouch™ *E. coli*/coliform count flexi plate supplied by HiMedia Laboratories, Pvt. Ltd. Mumbai, India.

2.3. Antibiotic susceptibility screening and MAR index

Disk diffusion method by Kirby-Bauer, was used to examine the antibiotic susceptibility profile of the isolates. The antibiotics used in the test belonged to ten different antimicrobial groups: 1) beta-lactams: penicillin (P 30 µg), amoxicillin (AMX 30 µg); 2)

Table 1
Physicochemical and bacteriological characteristics of river Ganga collected from different districts of Uttar Pradesh.

Sites	Coordinates	Physico-chemical Parameters						Bacteriological Parameters	
		pH	Color (Hazen Units)	DO (mg/L)	TSS (mg/L)	BOD (mg/L)	COD (mg/L)	Total Coliforms (MPN/100 mL)	Fecal Coliforms (MPN/100 mL)
Kanpur	26.4499° N, 80.3319° E	6.75 $\pm$ 0.11	32.04 $\pm$ 4.12	1.07 $\pm$ 0.21	86.33 $\pm$ 6.43	32.33 $\pm$ 3.05	90.32 $\pm$ 3.29	66000.00 $\pm$ 2000.00	32333.33 $\pm$ 1527.52
Dalmau	26.0682° N, 81.0617° E	7.22 $\pm$ 0.28	15.49 $\pm$ 5.07	3.95 $\pm$ 0.18	82.56 $\pm$ 3.83	57.73 $\pm$ 2.61	91.74 $\pm$ 2.04	70666.67 $\pm$ 2081.66	31333.33 $\pm$ 2516.61
Kunda	25.7175° N, 81.5212° E	8.35 $\pm$ 0.46	35.39 $\pm$ 3.19	5.66 $\pm$ 0.37	85.00 $\pm$ 11.53	29.00 $\pm$ 1.24	61.68 $\pm$ 3.85	70333.33 $\pm$ 2081.66	36666.67 $\pm$ 1527.52
Manikpur	25.0699° N, 81.1031° E	7.50 $\pm$ 0.31	30.98 $\pm$ 4.98	4.93 $\pm$ 0.33	97.00 $\pm$ 8.00	32.56 $\pm$ 2.07	95.63 $\pm$ 3.05	72666.67 $\pm$ 2081.66	30666.67 $\pm$ 2516.61
Jhunsu	25.4290° N, 81.9140° E	6.85 $\pm$ 0.46	49.17 $\pm$ 3.71	3.67 $\pm$ 0.55	89.33 $\pm$ 6.11	31.03 $\pm$ 3.01	80.93 $\pm$ 3.23	70333.33 $\pm$ 6027.71	38000.00 $\pm$ 3605.55
Mirzapur	25.1337° N, 82.5644° E	7.20 $\pm$ 0.57	34.28 $\pm$ 3.80	3.62 $\pm$ 0.59	70.90 $\pm$ 6.77	27.73 $\pm$ 1.52	91.30 $\pm$ 8.70	71333.33 $\pm$ 7023.76	32000.00 $\pm$ 5567.76
Varanasi	25.3176° N, 82.9739° E	6.71 $\pm$ 0.61	32.93 $\pm$ 4.02	2.94 $\pm$ 0.71	72.66 $\pm$ 9.61	31.33 $\pm$ 3.51	79.23 $\pm$ 7.85	68666.67 $\pm$ 3511.88	32666.67 $\pm$ 5507.57
Ramnagar	25.2764° N, 83.0334° E	6.73 $\pm$ 0.35	43.79 $\pm$ 3.26	3.28 $\pm$ 0.51	69.33 $\pm$ 4.00	28.46 $\pm$ 2.45	78.30 $\pm$ 7.49	64333.33 $\pm$ 4041.45	28333.33 $\pm$ 3055.05
Mughalsarai	25.2838° N, 83.1156° E	6.75 $\pm$ 0.45	44.70 $\pm$ 3.12	3.97 $\pm$ 0.55	63.33 $\pm$ 10.01	28.56 $\pm$ 2.90	75.96 $\pm$ 6.88	63000.00 $\pm$ 5567.76	29000.00 $\pm$ 7211.10
Saidpur	25.5518° N, 83.1834° E	7.06 $\pm$ 0.33	51.59 $\pm$ 1.67	3.77 $\pm$ 0.73	77.66 $\pm$ 6.50	28.86 $\pm$ 3.28	88.66 $\pm$ 6.50	72333.33 $\pm$ 4163.33	32333.33 $\pm$ 4509.25
Ballia	25.8307° N, 84.1857° E	7.16 $\pm$ 0.50	25.42 $\pm$ 1.86	3.50 $\pm$ 0.47	55.66 $\pm$ 6.80	27.73 $\pm$ 2.76	82.80 $\pm$ 7.11	73333.33 $\pm$ 6027.71	32666.67 $\pm$ 4509.25
Drinking water quality standard (IS 10500:2012)		6.5–8.5	5	NS	NS	NS	NS	0	0

*NS= Not Specified.

Analysis was performed in triplicates. All values are mean $\pm$ SD.

cephalosporin: cefazolin (CZ 30 µg), cefuroxime (CXM 30 µg); fluoroquinolones: norfloxacin (NX 10 µg), ofloxacin (OX 5 µg); aminoglycosides: streptomycin (S 10 µg), gentamicin (GEN10µg); carbapenems: imipenem (IMP750µg); tetracycline: tetracycline (TE30µg); macrolides-erythromycin (E 15 µg); lincosamides: clindamycin (CD 2 µg); sulphonamides: co-trimoxazole (COT25µg), trimethoprim (TR25µg) and glycopeptides: teicoplanin (TEI 30 µg), vancomycin (VAN 30 µg). The results were interpreted according to CLSI breakpoints (CLSI, 2017). Isolates that displayed resistance to 3 or more different antibiotics classes were classified as multidrug resistant (MDR) (Magiorakos et al., 2012; Smyth et al., 2020). The percentage of resistant and sensitive isolates was calculated as: percentage (%) = [(Number of resistant OR sensitive isolate to an antibiotic/total number of tested isolates) X 100] (Smyth et al., 2020). On the basis of the resistance phenotype obtained, the MAR index of the sampling sites was calculated according to Kruperman (1983).

2.4. Determination of ESBL and AmpC phenotype

The isolates were phenotypically tested for the production of beta-lactamase enzymes such as ESBL and AmpC. The double disk synergy test using ceftazidime (CAZ 30 µg), cefotaxime (CTX 30 µg) and an augmentin- AMC (20 µg Amoxicillin and 10 µg Clavulanic acid) disk was done to check for the presence of ESBL production (CLSI, 2013). An increase in the inhibition zone of the cephalosporin disk in the direction of the augmentin disk indicates positive result. For AmpC production antibiotics Cefoxitin (CX 30 µg) and Cefoxitin-Cloxacillin (CXX 200 µg) were used. An increase in the zone diameter of ≥ 4 mm in the CXX zone than CX alone was considered as a positive result (Bajaj et al., 2015).

2.5. Molecular characterization of beta-lactamase genes

Genomic DNA was extracted from overnight grown culture of

isolates positive for MDR, ESBL and AmpC following the alkaline lysis method. Multiplex PCR was performed to confirm the presence of ESBL genotypes: *bla*TEM, *bla*SHV, *bla*OXA and *bla*CTX-M (CTX-M-1, CTX-M-2 and CTX-M-9). The thermocycling conditions were described as: initial denaturation at 94 °C for 10 min followed by 30 cycles of denaturation at 94 °C for 40s, annealing at 60 °C for 40s and extension at 72 °C for 1min; and a final elongation step at 72 °C for 7min (Dallenne et al., 2010). Similarly the genes encoding enzymes for AmpC β-lactamases such as MOX, CIT, DHA, ACC, EBC and FOX were also studied by multiplex PCR in the phenotypic positive isolates. AmpC genes were detected by multiplex PCR using conditions as described by Pérez-Pérez and Hanson (2002). The primer sequences and the product size (base pair) are summarized in Table 2. The amplified products were visualized under UV transilluminator (G: BOX Syngene, USA) after agarose gel electrophoresis. A 100-bp and 1-kb DNA ladder (New England Biolabs, USA) was used as a marker. To confirm the specific type of β-lactamase gene, nucleotides and its deduced amino acid sequences were compared with the available database on GeneBank (<http://www.lahey.org/Studies>).

2.6. Identification of isolates positive for MDR, ESBL and AmpC

The isolates positive for all the three traits i.e., displayed resistance to three or more antibiotics belonging to different groups and exhibited ESBL and AmpC genotypes were further identified by Sanger sequencing of the V3 and V4 region of bacterial 16S rRNA gene. The detection of 16S rRNA gene was done by using a set of universal primers (27F-AGAGTTTGATCCTGGCTCAG and 1492R-TACGGTTACCTTGTACGACTT). PCR products were further purified using quick gel extraction kit (Invitrogen, ThermoFisher Scientific, US). The amplified products were sequenced and the sequences were analyzed and compared with those available in GenBank, using the BLAST program (Basic Local Alignment Search Tool) from the National Center for Biotechnology Information of the United

Table 2
Details of the primer used to detect ESBL and AmpC genes.

Primer name	Target gene	Primer sequence (5'-3')	Amplicon size (bp)	Reference
Primers for ESBL genes				
TEM (F)	blaTEM	F-CAT TTC CGT GTC GCC CTT ATT C	800	Dallenne et al. (2010)
TEM (R)		R-CGT TCA TCC ATA GTT GCC TGA C		
SHV (F)	blaSHV	F-AGC CGC TTG AGC AAA TTA AAC	713	
SHV (R)		R-ATC CCG CAG ATA AAT CAC CAC		
OXA (F)	blaOXA	F-GGC ACC AGA TTC AAC TTT CAA G	564	
OXA (R)		R-GAC CCC AAG TTT CCT GTA AGTG		
CTX-M-1(F)	blaCTX-M-1	F-TTA GGA ART GTG CCG CTG YA	688	
CTX-M-1(R)		R-CGA TAT CGT TGG TGG TRC CAT		
CTX-M-2(F)	blaCTX-M-2	F-CGT TAA CGG CAC GAT GAC	404	
CTX-M-2(R)		R-CGA TAT CGT TGG TGG TRC CAT		
CTX-M-9(F)	blaCTX-M-9	F-TCA AGC CTG CCG ATC TGG T	561	
CTX-M-9(R)		R-TGA TTC TCG CCG CTG AAG		
Primers for AmpC genes				
MOXM-F	MOX	F-GCT GCT CAA GGA GCA CAG GAT	520	Pérez-Pérez and Hanson (2002)
MOXM-R		R-CAC ATT GAC ATA GGT GTG GTG C		
CITM-F	CIT	F-TGG CCA GAA CTG ACA GGC AAA	462	
CITM-R		R-TTT CTC CTG AAC GTG GCT GGC		
DHAM-F	DHA	F-AAC TTT CAC AGG TGT GCT GGG T	405	
DHAM-R		R-CCG TAC GCA TAC TGG CTT TGC		
ACCM-F	ACC	F-AAC AGC CTC AGC AGC CGG TTA	346	
ACCM-R		R-TTC GCC GCA ATC ATC CCT AGC		
EBCM-F	EBC	F-TCG GTA AAG CCG ATG TTG CGG	302	
EBCM-R		R-CTT CCA CTG CGG CTG CCA GTT		
FOXM-F	FOX	F-AAC ATG GGG TAT CAG GGA GAT G	190	
FOXM-R		R-CAA AGC GCG TAA CCG GAT TGG		

Table 3

Occurrence of beta-lactamase genes and antibiotic resistance profile of MDR isolates recovered from various sites.

Sites	Isolates	ESBL/AmpC Genotype	Resistance Phenotype	No. of resistant classes
Kanpur	IITR-AT-A1	<i>bla</i> TEM + <i>bla</i> SHV	AMX, P, CZ, CXM, COT, TR, VA, TEI	4
	IITR-AT-A2	<i>bla</i> TEM + <i>bla</i> SHV	AMX, P, CZ, CD, TEI	4
	IITR-AT-A3	<i>bla</i> TEM + <i>bla</i> SHV	AMX, S, TE, CD, VA, TEI	5
	IITR-AT-A7	<i>bla</i> TEM	P, GEN, CD, TR, VA, TEI	5
	IITR-AT-A10	<i>bla</i> TEM + <i>bla</i> CTX-M-1	NX, CD, TEI	3
	IITR-AT-A11	<i>bla</i> TEM + <i>bla</i> CTX-M-1 + <i>bla</i> CTX-M-9	AMX, P, CZ, CD, COT, TR, TEI	5
	IITR-AT-A12	<i>bla</i> TEM/CIT	P, CZ, CXM, TE, CD, COT, TR, TEI	6
	IITR-AT-A13	<i>bla</i> TEM	AMX, P, CZ, GEN, CD, TEI	5
	IITR-AT-A14	<i>bla</i> TEM	AMX, P, CZ, CD	3
	IITR-AT-B2	ND	AMX, P, CD, TEI	3
	IITR-AT-B3	<i>bla</i> TEM	P, CZ, CD, COT, TEI	5
Dalmau	IITR-AT-B4	<i>bla</i> TEM	AMX, P, CXM, CD, TEI	4
	IITR-AT-B5	<i>bla</i> TEM	AMX, CZ, CD, TEI	4
	IITR-AT-B6	<i>bla</i> TEM	AMX, P, CXM, E, CD, TEI	5
	IITR-AT-B7	<i>bla</i> TEM + <i>bla</i> CTX-M-9	CZ, CD, TEI	3
	IITR-AT-B8	<i>bla</i> TEM	AMX, P, CZ, CD, COT, TEI	5
	IITR-AT-B10	<i>bla</i> TEM	AMX, P, CXM, CD, COT, VA, TEI	5
	IITR-AT-B11	<i>bla</i> TEM	AMX, CD, TEI	3
	IITR-AT-C3	ND	P, IPM, CD	3
	IITR-AT-C5	ND	AMX, P, CD, TEI	3
	IITR-AT-D1	<i>bla</i> TEM	AMX, P, CZ, CXM, CD, COT, TR, VA, TEI	5
	IITR-AT-D2	<i>bla</i> TEM/MOX, DHA	AMX, P, CXM, NX, CD, COT, TR, TEI	6
Kunda	IITR-AT-D3	ND	AMX, P, CZ, CD, TEI	4
	IITR-AT-D4	<i>bla</i> TEM + <i>bla</i> CTX-M-2	AMX, P, CZ, E, CD, TEI	5
	IITR-AT-D6	ND	AMX, P, CZ, CXM, CD, VA, TEI	4
	IITR-AT-D7	<i>bla</i> TEM	P, CD, VA, TEI	3
	IITR-AT-E1	<i>bla</i> TEM/ACC	AMX, P, CZ, S, CD, COT	5
	IITR-AT-E2	ND	AMX, P, CZ, NX, S, CD, COT	6
	IITR-AT-E3	<i>bla</i> CTX-M-2	AMX, P, CZ, S, CD, COT	5
	IITR-AT-E4	<i>bla</i> CTX-M-1	AMX, P, CZ, S, CD, COT	5
	IITR-AT-E5	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	P, CZ, CXM, CD, TEI	4
	IITR-AT-E6	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA/CIT	P, CD, COT, TEI	4
	IITR-AT-E7	<i>bla</i> TEM	AMX, P, CXM, TR	3
Jhunsu	IITR-AT-E8	<i>bla</i> TEM	AMX, P, CZ, CD, COT, TEI	5
	IITR-AT-E9	ND	P, CZ, TEI	3
	IITR-AT-E10	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	CZ, E, CD, COT, TEI	5
	IITR-AT-E11	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA/FOX, ACC	AMX, P, CD, VA, TEI	3
	IITR-AT-E12	ND	AMX, P, CZ, CXM, E, CD, COT, TR, VA, TEI	6
	IITR-AT-E13	<i>bla</i> TEM + <i>bla</i> SHV	P, CD, NX, VA, TEI	4
	IITR-AT-E14	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	AMX, P, CD, TR, TEI	4
	IITR-AT-E15	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	AMX, P, CD, TEI	3
	IITR-AT-E16	ND	AMX, P, CZ, CD, TEI	4
	IITR-AT-E17	<i>bla</i> TEM + <i>bla</i> SHV	P, CZ, CD, TEI	4
	IITR-AT-E18	<i>bla</i> TEM + <i>bla</i> SHV	AMX, P, CZ, CD, TEI	4
Mirzapur	IITR-AT-E19	<i>bla</i> TEM	AMX, P, CZ, E, CD, COT, TR	5
	IITR-AT-E20	<i>bla</i> TEM	AMX, P, CD, TEI	3
	IITR-AT-E21	<i>bla</i> TEM + <i>bla</i> SHV	AMX, P, CD, VA, TEI	3
	IITR-AT-E22	ND	AMX, P, CZ, E, CD, COT, TR	5
	IITR-AT-E23	<i>bla</i> TEM + <i>bla</i> SHV	AMX, P, CD, VA, TEI	3
	IITR-AT-E24	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	AMX, P, CD, TEI	3
	IITR-AT-E25	<i>bla</i> TEM	P, CZ, CD, VA, TEI	4
	IITR-AT-E26	ND	P, CZ, CD, TEI	4
	IITR-AT-E27	<i>bla</i> TEM	P, CD, TEI	3
	IITR-AT-E28	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	AMX, P, CZ, CD, TEI	4
	IITR-AT-E29	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	AMX, P, CZ, CD, NX	4
Mirzapur	IITR-AT-E30	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	P, CZ, CD, TEI	4
	IITR-AT-E31	ND	AMX, P, CXM, TR	3
	IITR-AT-E32	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	AMX, P, CZ, CXM, CD, TEI	4
	IITR-AT-E33	ND	AMX, P, CXM, TR, VA	4
	IITR-AT-E34	<i>bla</i> TEM + <i>bla</i> SHV	AMX, P, CZ, IPM, CD, TEI	5
	IITR-AT-E35	ACC	AMX, P, CZ, CXM, IPM, TR	4
	IITR-AT-E38	ND	AMX, P, CD, TR, TEI	4
	IITR-AT-E39	ND	P, CZ, CXM, CD, VA, TEI	4
	IITR-AT-E40	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	AMX, P, TE, CD	3
	IITR-AT-E42	<i>bla</i> TEM + <i>bla</i> SHV	AMX, P, CZ, NX, TE, CD, COT, TR, TEI	6
	IITR-AT-E44	<i>bla</i> TEM	AMX, P, CZ, CD, TEI	4
Mirzapur	IITR-AT-E45	ND	AXM, P, CZ, OF, IPM, CD, TEI	6
	IITR-AT-F1	ND	AMX, P, CZ, CXM, COT, TR, TEI	4
	IITR-AT-F2	<i>bla</i> TEM + <i>bla</i> CTX-M-1	AMX, P, CZ, CXM, NX, CD, COT, TR, VA, TEI	6
	IITR-AT-F3	ND	AMX, P, CZ, IPM, CD, COT, TR, VA, TEI	6
	IITR-AT-F4	<i>bla</i> TEM	AMX, P, CZ, CD, TEI	4
	IITR-AT-F6	ND	P, CZ, VA, TEI	3
	IITR-AT-F7	ND	AMX, P, CZ, CXM, TE, CD, COT, TR, VA, TEI	6

(continued on next page)

Table 3 (continued)

Sites	Isolates	ESBL/AmpC Genotype	Resistance Phenotype	No. of resistant classes
Varanasi	IITR-AT-F8	ND	AMX, P, CZ, CXM, TE, E, CD, COT, TR, VA, TEI	7
	IITR-AT-F9	<i>bla</i> TEM/EBC, ACC	AMX, P, CZ, TE, E, CD, VA, TEI	6
	IITR-AT-F10	ND	AMX, P, CZ, CXM, E, CD, TEI	5
	IITR-AT-F11	ND	AMX, P, CZ, CXM, E, CD, COT, TR, TEI	6
	IITR-AT-F12	DHA, ACC	P, TE, CD, VA, TEI	4
	IITR-AT-F13	ND	CZ, CXM, CD, COT, TR, TEI	4
	IITR-AT-F14	<i>bla</i> TEM	AMX, P, CZ, CD, COT, TR, TEI	5
	IITR-AT-F15	DHA, ACC	P, CZ, VA, TEI	3
	IITR-AT-F17	<i>bla</i> TEM	AMX, P, CZ, CD, VA, TEI	4
	IITR-AT-F18	<i>bla</i> TEM	AMX, P, CZ, CD, TEI	4
	IITR-AT-F19	<i>bla</i> TEM	AMX, P, CZ, CD, TEI	4
	IITR-AT-F20	<i>bla</i> TEM	AMX, P, CD, TEI	3
	IITR-AT-F21	<i>bla</i> TEM	AMX, P, CZ, CD, TEI	4
	IITR-AT-F24	ND	AMX, P, CZ, CXM, TE, CD, COT, VA, TEI	6
	IITR-AT-F26	<i>bla</i> TEM	CZ, CD, TR, TEI	4
	IITR-AT-F27	ND	AMX, P, CZ, CD, TEI	4
	IITR-AT-F28	ND	AMX, P, CZ, CXM, TE, CD, COT, TR, TEI	6
	IITR-AT-F32	<i>bla</i> TEM	P, CZ, CXM, COT, TEI	4
	IITR-AT-F33	<i>bla</i> TEM	AMX, P, CZ, CD, TEI	4
	IITR-AT-F34	<i>bla</i> TEM	AMX, P, CZ, CD, TEI	4
	IITR-AT-F35	<i>bla</i> TEM	AMX, CZ, CD, VA, TEI	4
	IITR-AT-F36	<i>bla</i> TEM	AMX, P, CZ, CD, TEI	4
	IITR-AT-F37	<i>bla</i> TEM	AMX, P, CZ, CXM, CD, TEI	4
	IITR-AT-F38	<i>bla</i> TEM	AMX, P, CZ, TE, CD, VA, TEI	5
	IITR-AT-G2	<i>bla</i> TEM	AMX, P, CZ, CXM, E, CD, TEI	5
	IITR-AT-G3	ND	P, CZ, CD, TEI	4
	IITR-AT-G4	<i>bla</i> TEM + <i>bla</i> SHV	P, CZ, CXM, S, CD, COT, VA, TEI	6
	IITR-AT-G5	ND	P, CXM, CD, COT, TEI	5
	IITR-AT-G6	ND	AMX, P, CZ, CXM, CD, COT, TEI	5
	IITR-AT-G7	<i>bla</i> TEM	AMX, P, CZ, CXM, CD, TEI	4
	IITR-AT-G8	<i>bla</i> TEM	AMX, P, CZ, CD, COT, TR, VA, TEI	5
	IITR-AT-G9	ND	AXM, P, CZ, CXM, NX, OF, GEN, S, IPM, TE, E, CD, COT, VA, TEI	10
	IITR-AT-G10	ND	AXM, P, CZ, CXM, NX, OF, TE, CD, COT, TR, VA	7
	IITR-AT-G11	ND	AMX, P, CZ, CXM, CD, TEI	4
Ramnagar	IITR-AT-H1	<i>bla</i> TEM + <i>bla</i> CTX-M-1 + <i>bla</i> CTX-M-9	AMX, COT, TEI	3
	IITR-AT-H2	<i>bla</i> TEM	CZ, COT, TEI	3
	IITR-AT-H3	ND	AMX, CD, VA, TEI	3
	IITR-AT-H4	ND	AMX, CD, VA, TEI	3
	IITR-AT-H6	ND	AMX, CZ, CXM, E, CD, COT, TEI	6
	IITR-AT-H7	ND	AMX, CZ, CD, COT, TEI	5
	IITR-AT-H8	<i>bla</i> CTX-M-9	AMX, CZ, NX, COT, TEI	5
	IITR-AT-H9	ND	AMX, CZ, CXM, E, CD, VA, TEI	5
	IITR-AT-H10	ND	AMX, P, CZ, CXM, CD, COT, TEI	5
	IITR-AT-H11	ND	AMX, CZ, CXM, COT, TEI	4
	IITR-AT-H12	ND	AMX, CZ, CXM, E, COT, VA, TEI	5
	IITR-AT-H14	<i>bla</i> TEM	AMX, CZ, CD, COT, TEI	5
	IITR-AT-H15	ND	AMX, P, CZ, E, CD, COT, TEI	6
	IITR-AT-H16	ND	AMX, P, CZ, CD, COT, TEI	5
	IITR-AT-H17	ND	AMX, P, CZ, CD, COT, TEI	5
	IITR-AT-H18	ND	AMX, P, CZ, CXM, E, CD, COT, TR, TEI	6
	IITR-AT-H19	ND	AMX, P, CZ, CD, COT, TEI	5
	IITR-AT-H20	<i>bla</i> TEM	AMX, P, CZ, E, CD, COT, TR, TEI	5
	IITR-AT-H21	ND	AMX, P, CZ, CD, COT, TEI	5
Mughalsarai	IITR-AT-H23	ND	AMX, P, CZ, CD, COT, TEI	5
	IITR-AT-H24	ND	AMX, P, CZ, CXM, COT, TR	3
	IITR-AT-I1	<i>bla</i> TEM	AMX, P, CZ, CXM, E, CD, VA, TEI	5
	IITR-AT-I2	<i>bla</i> TEM	AMX, P, CZ, CXM, NX, E, CD, VA, TEI	6
	IITR-AT-I3	ND	AMX, P, CZ, CXM, NX, E, CD, VA, TEI	6
	IITR-AT-I4	ND	AMX, P, CZ, CXM, E, CD, VA, TEI	5
	IITR-AT-I5	ND	AMX, P, CZ, CXM, E, CD, VA, TEI	5
	IITR-AT-I6	FOX, CIT, ACC	AMX, P, CZ, E, CD, COT, VA, TEI	6
	IITR-AT-I8	<i>bla</i> TEM	AMX, P, CZ, CXM, E, CD, COT, VA, TEI	6
	IITR-AT-I9	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	AMX, P, CZ, CD, VA, TEI	4
	IITR-AT-I11	ND	AMX, P, CZ, CD, TEI	4
	IITR-AT-J51	ND	AMX, P, CZ, CD, TEI	4
	IITR-AT-J52	<i>bla</i> TEM	AMX, P, CZ, CD, COT, TEI	5
	IITR-AT-J53	<i>bla</i> TEM	AMX, P, CXM, CD, COT, TEI	5
	IITR-AT-J54	ND	AMX, P, CZ, E, CD, TEI	5
	IITR-AT-J55	ND	AMX, P, CD, VA, TEI	3
	IITR-AT-J56	ND	AMX, P, CZ, CD, VA, TEI	4
	IITR-AT-J57	ND	P, CZ, CXM, CD, COT, TR, VA, TEI	5
	IITR-AT-J58	ND	AMX, P, CZ, CXM, S, E	4
	IITR-AT-J59	ND	AMX, P, CZ, NX, CD, COT, TR, VA, TEI	6
	IITR-AT-J60	ND	AMX, P, CZ, CXM, E, CD, COT, TR, TEI	6
	IITR-AT-J61	ND	AMX, P, CZ, CXM, E, CD, COT, VA, TEI	6

Table 3 (continued)

Sites	Isolates	ESBL/AmpC Genotype	Resistance Phenotype	No. of resistant classes
Ballia	IITR-AT-J62	ND	AMX, P, CZ, CD, VA, TEI	4
	IITR-AT-K2	ND	AMX, P, CZ, CXM, NX, OF, GEN, S, IPM, TE, CD, COT, TR, VA	9
	IITR-AT-K3	<i>blaSHV</i> + <i>blaCTX-M-2</i>	AMX, P, TE, CD, COT, TR, TEI	5
	IITR-AT-K4	ND	AMX, P, CZ, E, CD, VA, TEI	5
	IITR-AT-K5	<i>blaTEM</i>	AMX, P, CZ, CD, VA, TEI	4
	IITR-AT-K6	<i>blaTEM</i>	AMX, P, CZ, CXM, NX, GEN, S, CD, COT, TR	6
	IITR-AT-K7	ND	AMX, P, CZ, E, CD, VA, TEI	5
	IITR-AT-K8	<i>blaTEM</i>	AMX, P, CZ, S, TE, CD, TEI	6
	IITR-AT-K9	<i>blaTEM</i>	AMX, P, CZ, E, CD, VA, TEI	5
	IITR-AT-K10	<i>blaTEM</i>	AMX, P, CZ, NX, S	4

AMX-Amoxicillin; P-Penicillin; CZ-Cefazolin; CXM-Cefuroxime; NX-Norfloxacin; OF-Ofloxacin; GEN-Gentamycin; S-Streptomycin; IPM-Imipenem; TE-Tetracycline; E-Erythromycin; CD-Clindamycin; COT-Co-trimoxazole; TR-Trimethoprim; VA-Vancomycin; TEI-Teicoplanin, ND- Not Detected, Resistance was defined as disk diameters above clinical break-point (CLSI).

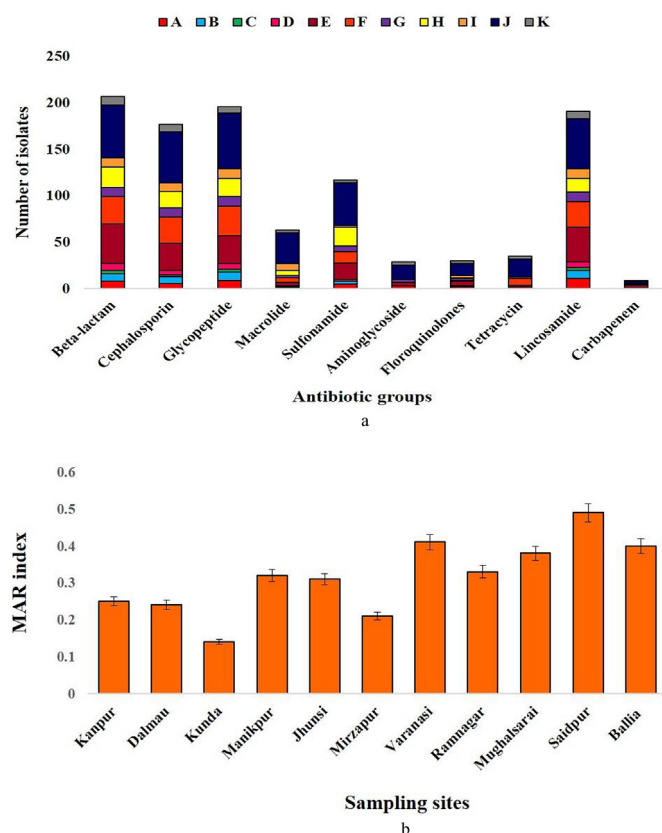


Fig. 2. a: Antibiotic resistance pattern observed at sampling stations located on banks of river Ganga in Uttar Pradesh, India (A: Kapur; B: Dalmau; C: Kunda; D: Manikpur; E: Jhansi; F: Mirzapur; G: Varanasi; H: Ramnagar; I: Mughalsarai; J: Saidpur; K: Ballia). b: Multiple Antibiotic Resistance (MAR) index of sampling sites on Ganga in Uttar Pradesh, India.

States (NCBI) (www.ncbi.nlm.nih.gov/BLAST).

3. Result

3.1. Water quality assessment of river Ganga

For qualitative analysis the water samples collected from river Ganga at different geographical indices was analyzed for its physico-chemical and bacteriological properties. The parameters that were tested include pH, color, DO, BOD, COD, TSS, Total coliforms and fecal coliforms. During the analysis it was found that the water quality not only varied greatly at different locations

which may be due to the seasonal and temporal conditions faced by the river but also exceeded the standards set by the regulatory agencies (Bureau of Indian Standards BIS, 2012). The results are mentioned in Table 1.

3.2. Resistance screening

In the present study, a total of 243 different bacterial colonies were isolated from 11 sites. Among 243 isolates, 177 (72.84%) isolates were found to be Gram-negative and 66 (27.16%) were Gram-positive. All 243 isolates were subjected to MDR screening to 16 antibiotics belonging to 10 different groups. A total of 206 (84.77%) isolates displayed the MDR trait i.e., was resistant to 3 or more antibiotics of different classes. Table 3 lists the resistance profile of the MDR isolates recovered from the sampling sites. Maximum resistance was observed against beta-lactams (P = 78.19%; AMX = 72.84%), glycopeptide (VAN = 32.92%; TEI = 79.42%), cephalosporin (CF = 67.90%; CFX = 38.27%), lincosamide (CD = 78.18%) followed by sulfonamide (COT = 43.20%; TRIM = 17.69%), macrolide (ERM = 26.33%) and tetracyclin (TET = 14.40%) (Fig. 2). The isolates displayed least resistance towards antibiotics belonging to groups-fluoroquinolones (NX = 11.52%; OF = 4.11%), aminoglycosides (GEN = 4.91%; S = 9.46%) and carbapenem (IMI = 3.70%) (Fig. 2a). Based on the resistance profiling of the MDR isolates, twelve resistant groups (Group I- Group XII) consisting of different combinations of antibiotics were formed (Table 4). During the analysis it was found that 21 (10.19%) isolates were showing resistance to maximum number of antibiotics used in the study.

The MAR index of maximum sites was found to be > 0.2, thus indicating high risk of antibiotic contamination and the role of such of sites in contaminating the river Ganga and the surrounding environment with pathogenic bacteria (Fig. 2b).

3.3. Characterization of beta-lactamase production

From 243 total isolates, 126 (51.85%) were confirmed as ESBL producers. *blaTEM* (95.24%) was the most prevalent ESBL gene that was detected in almost all of the ESBL positive isolates followed by *blaSHV* (22.22%), *blaOXA* (11.90%) and *blaCTX-M* (14.28%). Among, the 4 types of ESBL genes, *blaTEM* was found majorly alone (63.49%) followed by *blaCTX-M* (3.97%).

Some isolates harbored different combinations of ESBL genes (*blaTEM* + *blaSHV* = 11.11%; *blaTEM* + *blaCTX-M* = 8.73%; *blaTEM* + *blaSHV* + *blaOXA* = 10.32%). Prevalence of ESBL genotypes at various geographical indices is depicted in Fig. 3. Most of the ESBL positive isolates were found to harbor the MDR characteristic (82.54%) (Table 3). The observed ESBL-genotypes presented variation in the number of antibiotics the isolates were resistant to, with highest level of multi-drug resistance among *blaTEM*,

Table 4
Resistant groups identified in the MDR isolates.

Group No.	Resistant groups	Isolates harboring the groups	Sites displaying resistant groups
I	Beta-lactam, Cephalosporin, Lincosamides, Glycopeptides	30	Kanpur; Dalmau, Manikpur, Jhansi, Mirzapur, Varanasi, Mughalsarai, Saidpur, Ballia
II	Beta-lactam, Cephalosporin, Aminoglycosides, Lincosamides, Sulfonamides	2	Jhansi
III	Beta-lactam, Cephalosporin, Macrolides, Lincosamides, Glycopeptides	10	Manikpur, Mughalsarai, Mirzapur, Varanasi, Saidpur, Ballia
IV	Beta-lactam, Cephalosporin, Fluoroquinolones, Macrolides, Lincosamides, Glycopeptides	2	Mughalsarai
V	Beta-lactam, Cephalosporin, Lincosamides, Sulfonamides, Glycopeptides	17	Kanpur; Dalmau, Mirzapur, Varanasi, Ramnagar, Saidpur, Saidpur
VI a	Beta-lactam, Cephalosporin, Tetracycline, Macrolides, Lincosamides, Glycopeptides	3	
VI b	Beta-lactam, Cephalosporin, Tetracycline, Macrolides, Lincosamides, Sulfonamides, Glycopeptides	2	
VII a	Beta-lactam, Cephalosporin, Macrolides, Lincosamides, Sulfonamides, Glycopeptides	4	Saidpur, Mughalsarai
VII b	Beta-lactam, Cephalosporin, Macrolides, Lincosamides, Sulfonamides	2	Jhansi
VIII	Beta-lactam, Cephalosporin, Sulfonamides	2	Jhansi
IX	Beta-lactam, Lincosamides, Glycopeptides	12	Dalmau, Kunda, Manikpur, Jhansi, Varanasi, Ramnagar, Saidpur
X	Beta-lactam, Lincosamides, Sulfonamides, Glycopeptides	2	Jhansi
XI	Cephalosporin, Sulfonamides, Glycopeptides	2	Ramnagar, Saidpur
XII	Beta-lactam, Cephalosporin, Aminoglycosides, Fluoroquinolones, Carbapenem, Tetracycline, Macrolides, Lincosamides, Sulfonamides	2	Saidpur

Beta-lactam: Amoxicillin (AMX), Penicillin (P); Cephalosporin: Cefazolin (CZ), Cefuroxime (CMX); Fluoroquinolones: Norfloxacin (NX), Ofloxacin (OF); Aminoglycosides: Gentamycin (GEN), Streptomycin (S); Carbapenem: Imipenem (IPM); Tetracycline: Tetracycline (TE); Macrolides: Erythromycin (E); Lincosamides: Clindamycin (CD); Sulfonamides: Co-trimoxazole (COT), Trimethoprim (TR); Glycopeptides: Vancomycin (VA), Teicoplanin (TEI).

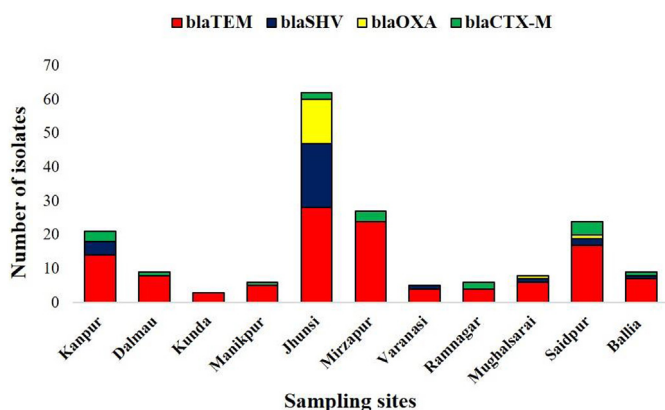


Fig. 3. Prevalence of ESBL genotypes at various geographical indices in Uttar Pradesh, India.

blaTEM + *blaSHV* and *blaTEM* + *blaSHV* + *blaOXA* (Fig. 4).

Plasmid mediated beta-lactamase enzyme AmpC was also detected in few isolates. Among 243, only 50 isolates were found positive for AmpC production when tested phenotypically. The genotypic analysis further reduced the number of AmpC positive isolates to 15 (6.17%). In comparison with ESBL, the percentage of AmpC producing isolates was very less. The AmpC genes that were identified in 15 isolates included MOX (6.67%), FOX (20.00%), CIT (33.33%), EBC (20.00%), ACC (46.67%) and DHA (26.67%). The distribution of ESBL and AmpC genes is depicted in Fig. 5.

3.4. Identification of the resistant isolates

The isolates that showed co-occurrence of the three resistant attributes i.e., MDR (3 or more antibiotic groups), ESBL and AmpC genotypes are designated as IITR-PC-A12; IITR-PC-E1; IITR-PC-E6 and IITR-PC-E11. On the basis of 16S rDNA similarity, the designated strains were identified as *Aeromonas caviae* (IITR-PC-A12), *Proteus vulgaris* (IITR-PC-E1), *Bacillus amyloliquefaciens* (IITR-PC-E6)

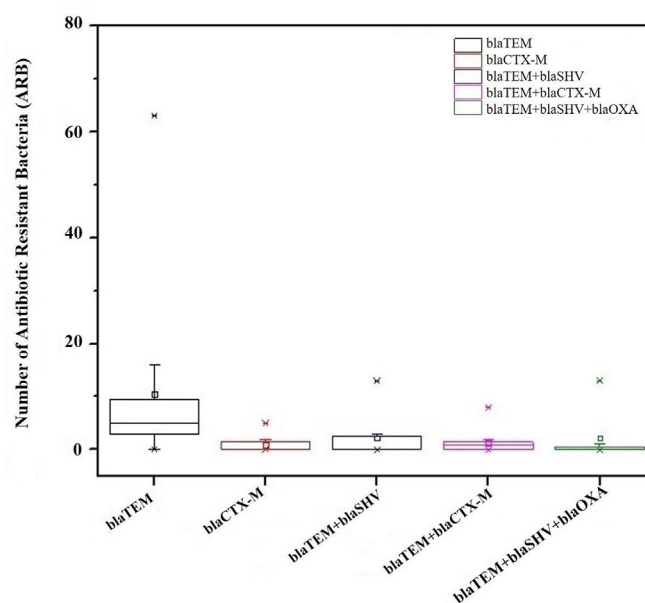


Fig. 4. Box plot showing presence of antimicrobial resistance (ARB) versus different ESBL genotypes in isolates. Solid horizontal line represents the median, the stems show the minimum and maximum values, and the cross indicates an outlier.

and *Enterobacter* species (IITR-PC-E11), respectively.

4. Discussion

Indiscriminate uses of antibiotics in domestic, commercial and agricultural sectors have exaggerated the antibiotic resistance threat in the environment (Korzeniewska et al., 2013; Osinaka et al., 2016). The effluents released from domestic household, WWTPs and industries are discharged into the aquatic environment without any adequate treatment. Moreover, the remedial steps offered at present for treatment of waste water at various WWTPs

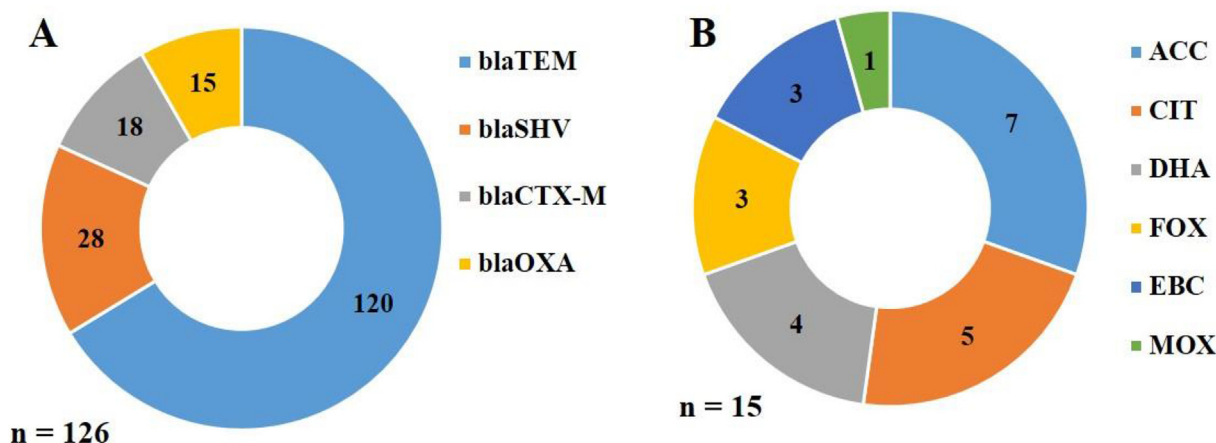


Fig. 5. Distribution of (A) ESBL and (B) AmpC genes in river Ganga, in Uttar Pradesh, India.

like activated sludge processes are inadequate for complete removal of chemical pollutants or genetic elements from the wastewater as it is centered on reduction or removal of the solid wastes (Suzuki et al., 2017). Polesel et al. (2016) described the factors and specific processes that influence the antibiotic removal in WWTPs like deconjugation, antibiotic retransformation of metabolites and transformation products, formation of analogues, hydrolysis of particulate as well as solid retention time. These technologies are cost effective and hence they are not deployed by developing countries like India. Moreover, the inadequate treatment generates an additional selection pressure on native bacterial strains in contaminated reservoirs which spreads the resistance in the environment (Pazda et al., 2019; Conte et al., 2017; Alouache et al., 2014). As a result of unmonitored discharge and inadequate treatment, the natural water bodies have turned into reservoirs of potent pathogenic bacteria harboring antibiotic resistance genes. The aquatic ecosystems further facilitates easy dispersal of resistant traits from clinical pathogens to environmental non-pathogenic strains (Karkman et al., 2018; Rizzo et al., 2013) through horizontal gene transfer (HGT) mediated by mobile genetic elements such as plasmids, integrons and transposons etc involving exchange of ARGs among different bacteria (Stange et al., 2019).

The current study is centered towards investigating the presence of bacterial strains depicting co-occurrence of multidrug resistance trait and genes for beta-lactamase production (ESBL, pAmpC) from river Ganga, within the region of Uttar Pradesh, India. In the present study, a total of 243 isolates were subjected to sensitivity test against antibiotics belonging to 10 different classes. Maximum resistance was recorded against antibiotics belonging to the classes-beta lactams, glycopeptides, cephalosporins, lincosamides followed by sulfonamides, macrolides and tetracyclins. As maximum number of isolates displayed resistance to drugs belonging to class-beta lactams and cephalosporin, the production of beta-lactamase enzymes in such isolates as a resistance mechanism becomes more obvious. Escalated resistance against beta-lactam antibiotics may be because of the fact that beta-lactam antibiotics are the highest consumed drugs in India (Van Boeckel et al., 2014). As the beta-lactam antibiotics are widely used in hospitals to treat the bacterial infections (Korzeniewska et al., 2013), the effluents released from these healthcare centers contain rich concentration of the drug which usually end up into the rivers through WWTPs. High incidence of resistance towards beta-lactam antibiotics has also been reported from several studies conducted using aquatic reservoirs as the source (Ye et al., 2017; Ferreira da Silva et al., 2007; Rodríguez-Mozaz et al., 2014). The

study also presented high resistance rates against glycopeptide and lincosamide drugs in the aquatic environment thereby introducing these classes as emerging resistant groups. Furthermore, resistance showcased against the classes macrolide, sulfonamide and tetracycline antibiotics supports the occurrence of resistant bacteria as these antibiotics are adequate for environmental monitoring of antibiotic resistance (Berendonk et al., 2015). Akiba et al. (2016), Rodríguez-Mozaz et al. (2014), Tacão et al., 2014, observed least resistance against antibiotics classes-fluoroquinolones, aminoglycosides and carbapenem, which depends on the factors like temperature, availability of nutrients brought about by effluent discharge and selection pressure etc. However, some findings also suggest that fluoroquinolones being the most dominant resistant group after with beta-lactams from environmental samples including rivers, hospital wastewater and WWTPs. Sulfonamides and fluoroquinolones are mainly used for treatment of urinary tract infections in humans whereas majority of the market is predominant with beta-lactam drugs, hence in our study while comparison with beta-lactams, lower resistance rate towards sulfonamides and fluoroquinolones were also noted as studied by Tennstedt et al. (2003); Pazda et al. (2019); Kotlarska et al. (2015). The reduced resistance rate for sulfonamide may be due to limited use (Pazda et al., 2019) in treatment of human infections. Although, usage of sulfonamide is seen in agriculture (Xu et al., 2016) and veterinary applications (Pazda et al., 2019) and hence its detection in surface runoff water may brought resistance.

As already reported by Pazda et al. (2019) the major groups of antibiotics that are mainly detected in wastewater are-beta-lactams, macrolides, quinolones, sulfonamides (& trimethoprim) and tetracyclins. As per recent report by Klein et al. (2018) India is one of the most antibiotic consuming countries and the consumption has been increased from 3.2 to 6.5 billion defined daily doses of antibiotics. As per the report there is 103% increase in consumption rate from 2000 to 2015. Similarly, in this study also the isolates depicted the same resistance traits against the suggested antibiotic groups, thereby illustrating their higher consumption rate in hospitals and community sectors leading to their unregulated discharge in wastewater and finally in rivers. The MAR index of ten sites out of 11 was found to be greater than 0.2 thus indicating high risk of antibiotic contamination at these sites (Kaushik et al., 2018).

As per the beta-lactamase characteristics, ESBL genes were detected in 51.85% isolates whereas plasmid-mediated AmpC genes were present in 6.17% isolates. Similar distribution of ESBL and AmpC producing isolates has also been reported previously (Bajaj

et al., 2015). In a similar study conducted in China to evaluate the presence of ESBL and pAmpC producing Enterobacteriaceae, the distribution of beta-lactamase genes was reverse i.e., ESBL⁺ were 11.6% and AmpC⁺ were 16.3% (Ye et al., 2017). The ESBL genes most commonly detected in this study were *bla*TEM followed by *bla*SHV, *bla*CTX-M and *bla*OXA. The wide distribution of *bla*TEM gene has also been reported in water samples collected from river Yamuna in New Delhi (Bajaj et al., 2015), from various water resources in Coimbatore (Natarajan and Subashkumar, 2017) and in effluents from WWTPs, Ireland (Smyth C. et al., 2020). High copy number of *bla*TEM gene has also been reported earlier in river Ganga by Singh et al. (2016). The continuous exposure to beta-lactam antibiotics may be responsible for creating selective pressure for *bla*TEM gene as these beta-lactam antibiotics are highly popular in the medical world due to their high clinical potency and reduced toxicity (Singh et al., 2016). The wide occurrence of *bla*TEM gene in the ESBL⁺ isolates is mainly due to the horizontal gene transfer through plasmids. However, this finding was in contrast with other studies in which *bla*CTX-M was most frequently reported (Ye et al., 2017; Akiba et al., 2016). The AmpC genes that were detected in the isolates included – MOX, FOX, CIT, EBC, ACC and DHA; gene ACC being the most prevalent. Earlier DHA-1 has also been reported as the predominant pAmpC genotype in water sample in China (Ye et al., 2017). The study further showed that ESBL producers presented MDR phenotype to other non beta-lactam antibiotics such as glycopeptides, lincosamides, macrolides, sulfonamides and quinolones. The co-resistance pattern of ESBL producers towards sulfonamides and quinolones has been identified in many previous studies (Conte et al., 2017; Yamashita et al., 2017; Tacão et al., 2014) but the present study has reported the occurrence of resistance to multiple antibiotics such as glycopeptides, lincosamides and macrolides in the ESBL resistant strains. It could be due to extreme selection pressure imposed by the effluents in the reservoir, thus acquiring resistance rates due to horizontal transfer might be higher; also plasmidic origin AmpC presence might have facilitated the multi drug resistance in surface water.

In the present study the isolates that exhibited the three resistant characteristics were identified as *Aeromonas caviae*, *Proteus vulgaris*, *Bacillus samyloliquefaciens* and *Enterobacter species*. The presence of *Aeromonas species* and its role in spreading the antibiotic resistance genes from wastewater to open environments has also been reported earlier (Piotrowska et al., 2017). A study in Nigeria has described the presence of *Proteus* sp., *Bacillus* sp. and *Enterobacter* sp., as drug resistant pathogens in pharmaceutical wastewater (Obayiuwana et al., 2018). Furthermore, several studies have reported *Proteus* sp. (*Proteus vulgaris*) as potent multi-drug resistant organisms and as potent producers of beta-lactamase enzymes: ESBL and AmpC (Girlich et al., 2020). Similarly, the ability of *Enterobacter* sp. to co-produce ESBL and AmpC beta-lactamase enzymes and its resistance to other antibiotics has also been published in earlier studies (Ojer-Usoz et al., 2014; Amador et al., 2015).

The findings in the study highlighted the role of wastewater from various sources (domestic, industrial, hospital and agriculture) in disseminating this soon-to-be pandemic threat across the globe (Ben Said et al., 2016; Alouache et al., 2014). Such pathogenic bacterial strains exhibiting co-occurrence of MDR, ESBL and AmpC traits are released into the receiving aquatic bodies like lakes and rivers and further proliferate and spread in the open environment (Mokracka et al., 2012; Ma et al., 2013). The river habitat impacted by several anthropogenic activities (overuse and misuse of antibiotics in agricultural and veterinary, sewage disposal, unregulated waste management practices and wastewater treatment) may provide a suitable platform for the exchange of genetic material through horizontal gene transfer causing the spread of multi-

resistant strains (Tacão et al., 2014). Mobile genetic elements such as plasmids and integrons are well known agents in the microbial world for the transmission of such factors. They harbor genetic determinants of resistance to many antibiotics of different classes turning the recipient strain into potent drug-resistant bacteria. The present research work highlights their role in disseminating the resistance phenomenon in the environment. Thus effective technologies that are not only capable of reducing the microbial load in the wastewater but are further capable of reducing the resistant strains present in the effluents before they are drained out into the open environment should be designed and implemented.

4.1. Practical applications and future research prospective

The waste waters are the potent reservoirs of many pathogenic bacteria and viruses; the effluents released from hospitals, industries and domestic settings should be monitored before its release into the open environment for reuse. A mandatory and strict policy should be adopted for treatment and reuse of wastewater to avoid, the introduction of resistant strains into the food chain. Such data can be used by local governing bodies for alarming health care providers to guard against the prevalent resistance in environment thereby assisting in risk management during epidemics. It could also assist in global surveillance of antibiotic resistance and policies to curtail unnecessary antibiotic use. The basic information generated from ecological perspective could assist in studying the factors involved in the co-occurrence of the resistant attributes i.e., MDR (3 or more antibiotic groups), ESBL and AmpC together, to curb the global dissemination of ARB and in studying the mechanism for its reversal.

5. Conclusion

This study concluded that surface water harbors and disseminates isolates and genes responsible for multidrug resistant (MDR), extended spectrum beta lactamase (ESBL) and AmpC beta lactamase traits. The study detected the presence of 84.77% MDR positive isolates posing maximum resistant to drugs belonging to β -lactams, glycopeptides, cephalosporins and lincosamides followed by sulfonamide, macrolide and tetracycline in river Ganga. ESBL production was reported in 51.85% isolates with *bla*TEM (95.24%) being the most prevalent ESBL gene. Such resistant strains can easily get introduced into the food chain through water bodies resulting in spread of environment borne infections. In the study, the presence of pathogenic antibiotic-resistant strains in surface water ecosystem like river Ganga or aquatic reservoirs is alarming and requires stringent surveillance for setting up of immediate measures to prevent the dissemination of antibiotic resistance.

Credit author statement

Preeti Chaturvedi (PC) and Deepshi Chaurasia (DC): Methodology, experimentation work, Writing, Resources Conceptualization, Investigation, Methodology, Writing - Original draft preparation, Project administration, Funding acquisition. **Ashok Pandey (AP) and Pratima Gupta (PG):** Supervision Formal analysis, Writing - Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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